

AI Chatbot Module Documentation

Overview

The AI chatbot is integrated into the hackathon landing page to provide instant responses related to registration, tracks, rules, prizes, timelines, and submissions.

Architecture

Frontend (Next.js Chat UI) → API Route → OpenRouter API → LLM + Hackathon Knowledge Context → AI Response

Technologies Used

- Next.js / React
- TypeScript
- Tailwind CSS
- OpenRouter API
- Large Language Model (LLM)

Current Model

Currently, the chatbot uses the **GPT OSS 20B** model through OpenRouter because it provides:

- Good response quality
- Fast inference speed
- Lower API cost
- Suitable performance for hackathon FAQ/chatbot use cases

RAG (Retrieval-Augmented Generation)

A lightweight RAG-style approach is implemented by injecting predefined hackathon context/knowledge into the AI prompt before generating responses. This allows the chatbot to answer event-specific questions accurately.

API Key Setup

Store the API key inside:

```
OPENROUTER_API_KEY=your_api_key_here
```

File location:

`.env.local`

Changing the AI Model

To switch models, update the `model` field inside:

`src/app/api/chatbot/route.ts`

Example:

`model: "openai/gpt-oss-20b"`

You can replace it with other supported models such as:

- `openai/gpt-oss-120b`
- `meta-llama/llama-3.3-70b`
- `qwen/qwen3-32b`

Chatbot Component Location

Frontend chatbot UI is located at:

`src/components/chatbot/chatbot.tsx`

Backend API logic is located at:

`src/app/api/chatbot/route.ts`